

BSCDS47(B): PREDICTIVE ANALYTICS

BSc Data Science. IV Year VII Sem.

Teaching Scheme: TH: - 3 Hours/Week	Credit TH: 03	Examination Scheme: In Sem. Evaluation: 25 Marks Mid Sem. Exam: 25 Marks End Sem. Exam: 50 Marks Total : 100 Marks
Course Prerequisites: 1. Basic understanding of human anatomy and physiology 2. Familiarity with biological terminology and medical conditions 3. Introductory knowledge of statistics and data analysis 4. Exposure to basic machine learning and classification techniques		
Course Objective: 1. To know various predictive data analysis models and use analytical tools to solve. 2. To Solve real life business problems. 3. To understand basic forecasting techniques to predict future values. 4. To Analyze various domain data with case studies. 5. Apply computational approaches and data science tools for prediction.		
Learning Course Outcome: At the end of the course, students will be able to: CO1: To assess the suitability of Predictive models for effective business decisions.. CO2: To collect analyze and visualize data; thereby utilize it in decision making.. CO3: To enhance the skills on linear and logistic regression CO4: Evaluate disease patterns using real-world case studies and bioinformatics tools. CO5: To apply forecasting techniques in making effective business decisions		
Course Contents		
UNIT-I	Simple Regression Analysis	9 Hours
• Concept Fundamentals of Regression Analysis - • Requirements in Regression Model Building - Model Diagnostics - Interpretation of Regression results for Management Decision. • Multiple Regression Analysis: Concept - Significance of Multiple Regression Analysis - Structure of Model Estimation - Testing Rule of Multiple Regression Analysis		
UNIT-II	Non-Linear Regression Analysis:	9 Hours
• Concept - Types of Non-linear Regression Models - Model Transformation - Difference between Linear and Non-linear Regression Models. • Diagnostics of Regression Modelling: Model Diagnostics - Multicollinearity - Autocorrelation		
UNIT-III	Dummy modelling and Panel Data Model Dummy modeling	9 Hours
• Dummy independent, modelling-linear probability Model-Logit model-Probit model Mental and neurological disorders • Panel Data Model: Concept - Panel Data Models - Fixed Effects Model - Random Effects Model - Forms of Panel Data Models - Applications to use Panel Data Models.		
UNIT-IV	Forecasting and Machine Learning: Time Series Forecasting	9 Hours

• Concept - Data Interpretation - Data Reduction - Classification and Clustering Techniques - Association Rule Mining - Cause and Effect Model.		
UNIT-V	Data Mining and Simulation: Data Mining	9 Hours
• Concept - Data Interpretation - Data Reduction - Classification and Clustering Techniques - Association Rule Mining - Cause and Effect Model. • Simulation: Concept - Monte Carlo Simulation - Discriminant Event Simulation - Application ,Using Simulation.		
TEXTBOOKS: 1. Predictive Analytics: The Power to Predict Who Will Click, Buy, Lie, or Die" by Eric Siegel 2. Data Science for Business" by Foster Provost and Tom Fawcett 3. Gareth James' Daniela Witten Trevor Hastie Robert Tibshirani. An Introduction to Statistical Learning with Applications in R		